

back against the Perspex sheet.

Next, switch off the CFL bulb, arrange the sandwich and the table lamp. Switch on the CFL bulb while setting the timer for 10 minutes. After the time elapses, turn off the light and prepare for the next step.

You must be wondering why use a CFL bulb instead of a UV light source. There are two reasons: One, CFL bulbs also emit some UV light along with the visible light. Two, these are readily available. The text on the protective foil of these PCBs states that you can expose them either using UV light for a duration of 60 to 90 seconds or using a CFL bulb for six to ten minutes. That suggests that the quantity of UV light emitted by a CFL bulb in a time window of around 10 minutes is equal to that from a UV bulb in around 90 seconds.

Step 5: Developing

During the ten minutes you turn on the CFL bulb, prepare fresh 'developer' solution. To do this, take a photographic developing tray. If that's not available, use a plastic or glass container with a capacity of about 750ml and dimensions of around 15x20cm² (6x8 inches). Pour 200ml of water in the container. Add half a teaspoon of caustic soda (NaOH) flakes to the water and stir the solution until the flakes fully dissolve in the solution. Ensure that you do not exceed the caustic soda quantity. As caustic soda is a hazardous chemical, use protective gloves and spectacles. You may be able to purchase caustic soda online if you look for flakes of drain cleaner for making soap. Alternatively, try at the local departmental store.

Note that the sequence is critical. Never put caustic soda first and then add water, as it may cause spurting and even violent eruption. If you use caustic soda in powder form, the solution will be too weak to develop the PCB.



Fig. 23: (Top) CFL bulb, (middle) the setup, and (bottom) the exposing process

After ten minutes of exposing the PCB laminate, shut off the CFL bulb and, in a dark-room setting, carefully remove the glass and the transparency to remove the 'developed' PCB. Immerse the PCB in the developer solution in the container with the pre-sensitised side of the PCB facing upwards. Keep stirring the solution; preferably with a 1cm broad hobby paint brush. PCB tracks will soon start becoming visible as green lines.

It may take as much as ten minutes to develop the PCB completely; the time may vary depending on the strength or potency of the flakes and the resulting developer solu-

tion. Patience is the key here; do not be tempted to add more flakes to accelerate the development as it may ruin the PCB. Continue stirring the developer solution until the PCB looks like the one in Fig. 26. It is fully developed now.

Remove the PCB and rinse it under running water. Wipe the PCB with a tissue paper and let it dry. Carefully inspect the PCB with a magnifying lens. If there is an undesirable bridge formed by the hardened green developer, remove it carefully using a sharp object like a pin. If copper is exposed in an area where it should not be, touch it up using a permanent marker pen with a narrow nib. You are all set for the next step now.

Step 6: Etching

There are several options for etching or removing the exposed copper from the PCB. I had a lot of ferric chloride (FeCl_3) solution, so I didn't look beyond it. FeCl_3 is available from chemical shops in the form of crystals and liquid. Please use protective gloves and spectacles for this step. If you have ferric chloride in crystal form, prepare the etching solution by adding 200 grams of FeCl_3 crystals in one litre of distilled water. Be cautious while handling it and use either glass or plastic to stir. Ferric chloride should not come in contact with skin. Do not spill or throw it away here and there as it will leave ugly, brown colour stains.

Remove the developer solution from the container, wash it well and pour around 200ml of ferric chloride (etchant) in it. Immerse the developed PCB into the etchant and continuously shake it slowly to ensure that the solution is swished from the PCB surface and fresh solution replaces it. Green-coloured areas on the PCB resist the etching solution and the exposed copper gets dissolved in it. This takes around 10 minutes. Remove the PCB and rinse it under running wa-